15.095 Machine Learning Under a Modern Optimization Lens

Interpretable Machine Learning for Telemonitoring Parkinson's Disease:
Predicting Severity from Acoustic Biomarkers

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1 Introduction

Parkinson's disease (PD) is a progressive neurodegenerative condition that affects millions worldwide. A key challenge in PD care is continuous symptom tracking, as progression varies widely across individuals. Traditional in-clinic assessments such as the Unified Parkinson's Disease Rating Scale (UPDRS) are costly, time-consuming, and often inaccessible for frequent monitoring. Recent work suggests that vocal biomarkers, which represent measurable changes in speech, are strongly correlated with PD severity, and can be collected cheaply and noninvasively. Using the UCI Parkinson's Telemonitoring dataset, which contains repeated voice recordings and clinician-rated UPDRS scores, we investigate whether acoustic features and simple demographic variables can accurately predict total UPDRS, enabling scalable telemonitoring.

Our goal is to develop predictive models that are both accurate and interpretable for this clinical setting. We compare a range of methods, from linear models (OLS, LASSO and MIO-based sparse/robust regression) to tree-based machine learning models (CART, Random Forests, XGBoost) and Optimal Regression Trees (ORT) implemented in Interpretable AI (IAI). We focus on understanding the trade-off between predictive performance and interpretability across different engineered feature sets.

Overall, our best model is an ORT trained on a small holistic feature set derived from transformed acoustic and demographic variables, which achieves test error comparable to or better than the best linear and ensemble baselines. At the same time, it is also able to summarize its predictions as a small set of clinically interpretable decision rules for UPDRS telemonitoring.

2 Data

2.1 Data Overview

The dataset consists of 5,875 voice recording observations from 42 subjects with early-stage Parkinson's disease. For each observation, clinicians recorded two UPDRS scores - motor UPDRS and total UPDRS - along with demographic attributes (subject ID, age, sex), time since enrollment, and 19 acoustic biomarkers derived from sustained phonation of the vowel a. These acoustic features quantify aspects of phonatory stability (jitter and shimmer variants), breathiness (NHR, HNR), complexity of vocal fold dynamics (RPDE, DFA), and fundamental frequency variation (PPE).

2.2 Data Preprocessing

Since multiple recordings are nested within each patient, we perform a subject-level train/validation/test split to prevent information leakage. Specifically, we randomly select 30 subjects for training, 6 for validation, and 6 for testing, ensuring complete independence across sets. We treat total UPDRS as the outcome and use all acoustic measures plus demographic covariates as predictors. Most acoustic features vary by several orders of magnitude, so we standardize all continuous variables using statistics computed from the training set, while leaving the binary sex variable unscaled. Missing values are not present in this dataset, simplifying preprocessing.

To further characterize the predictors, we examine the pairwise correlation matrix of all candidate features (Figure 1). The acoustic features show the following:

- high multicollinearity (especially among jitter- and shimmer-related measures),
- large scale differences across feature groups, requiring normalization, and

- non-constant variability across UPDRS levels, with patients at higher severity showing both stronger and more dispersed acoustic changes (i.e., the relationship between features and UPDRS is clearly nonlinear and heteroskedastic).

These properties motivate both our normalization step and the use of flexible nonlinear models, as well as more parsimonious engineered feature sets to mitigate redundancy.

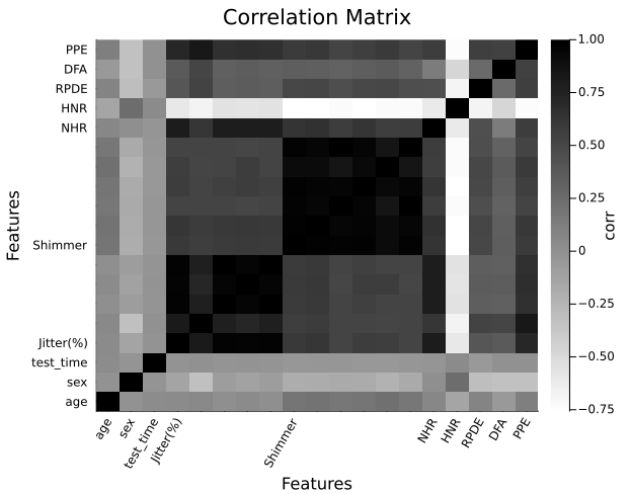

Figure 1: Correlation Matrix

By looking at the residuals of the features from an initial linear regression (Figure 2), we observe clear curved patterns and changes in spread for several acoustic variables. In other words, the errors are not randomly scattered around zero, which means the linear model does not fully capture how these features relate to UPDRS. This motivates us to apply nonlinear transformations and build the holistic regression model introduced in the next section.

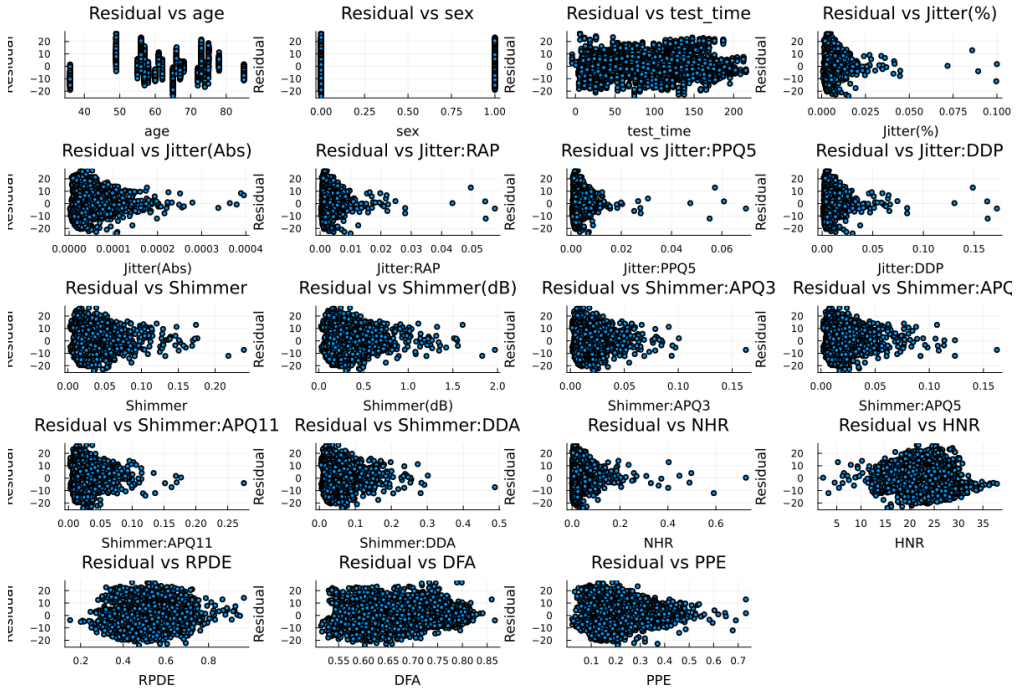

Figure 2: Residual Plots

3 Methods

Our analysis proceeded in stages, deliberately selecting features that satisfy optimization-based constraints before moving to the final tree models. We evaluated models along two key dimensions: predictive accuracy (measured by MAE and RMSE) and interpretability (measured by sparsity and coefficient stability).

3.1 Regression Models

3.1.1 Baseline Models

Ordinary Least Squares (OLS): A full OLS regression using all 19 predictors served as the baseline. The model minimizes the residual sum of squares:

$$\min_{\beta_0,\beta} \sum_{i=1}^{n} (y_i - \beta_0 - \mathbf{x}_i^\top \boldsymbol{\beta})^2 \tag{1}$$

This approach demonstrated poor stability and high test RMSE due to severe multicollinearity among acoustic features. With 19 correlated predictors and only 30 training subjects, the model suffered from overfitting and unreliable coefficient estimates, confirming the need for regularization or feature selection.

Lasso Regression: To address multicollinearity, we applied Lasso regression, which adds an ℓ_1 penalty to the objective function:

$$\min_{\beta_0,\beta} \sum_{i=1}^{n} (y_i - \beta_0 - \mathbf{x}_i^\top \boldsymbol{\beta})^2 + \lambda \sum_{j=1}^{p} |\beta_j| \tag{2}$$

where λ controls the regularization strength. The ℓ_1 penalty naturally performs feature selection by shrinking less important coefficients exactly to zero. We selected λ via 10-fold cross-validation on the combined training and validation sets.

Lasso significantly improved predictive accuracy over OLS, achieving a lower test RMSE and test MAE. It automatically selected 16 features, dropping redundant variables such as ‘Shimmer‘, ‘Shimmer:APQ3‘, and ‘Shimmer:DDA‘. However, Lasso retained a relatively large feature set, highlighting its primary limitation: it prioritizes predictive accuracy over maximum sparsity and may select multiple correlated features from the same group.

OLS with Lasso-Selected Features: We refitted an unpenalized OLS model using only the 16 features selected by Lasso. This approach achieved a test RMSE and test MAE slightly worse than Lasso itself, confirming that the shrinkage effect of the ℓ_1 penalty provides additional regularization beyond feature selection alone. Nevertheless, coefficient stability improved substantially compared to full OLS.

3.1.2 MIO Sparse Regression

To overcome Lasso's heuristic approach to feature selection, we implemented Mixed-Integer Optimization (MIO), which provides a globally optimal solution to the best subset selection problem. The MIO formulation explicitly selects exactly k features by solving:

$$\begin{aligned} \min_{\beta_0,\beta,z} \quad & \sum_{i=1}^{n} (y_i - \beta_0 - \mathbf{x}_i^\top \boldsymbol{\beta})^2 \\ \text{s.t.} \quad & \sum_{j=1}^{p} z_j \le k \\ & -M z_j \le \beta_j \le M z_j, \quad \forall j \\ & z_j \in \{0,1\}, \quad \forall j \end{aligned} \tag{3}$$

where z_j is a binary indicator (1 if feature j is selected, 0 otherwise), k is the desired sparsity level, and M is a sufficiently large constant (set to 1000 in our experiments). The "Big-M" constraints ensure that $\beta_j = 0$ whenever $z_j = 0$.

We evaluated sparsity levels $k \in \{5, 6, 7, 8, 9, 10\}$ to explore the accuracy–sparsity trade-off. The optimal MIO model selected $k = 9$ features based on RMSE and $k = 10$ features based on MAE, outperforming Lasso

at comparable sparsity levels. Unlike Lasso, MIO guarantees the optimal k-sparse solution, finding feature combinations that Lasso's greedy path might miss. Compared to Lasso, MIO achieved better predictive performance for MAE with fewer features, demonstrating the value of global optimization for subset selection. However, MIO's computational cost was significantly higher than that of Lasso's.

3.1.3 MIO Robust Regression with Sparsity

Building on MIO sparse regression, we added an ℓ_1 robustness penalty to combine the benefits of optimal feature selection with coefficient shrinkage:

$$\begin{aligned} \min_{\beta_0,\beta,z} \quad & \sum_{i=1}^{n}(y_i - \beta_0 - \mathbf{x}_i^\top \boldsymbol{\beta})^2 + \Gamma \sum_{j=1}^{p} |\beta_j| \\ \text{s.t.} \quad & \sum_{j=1}^{p} z_j \leq k \\ & -Mz_j \leq \beta_j \leq Mz_j, \quad \forall j \\ & z_j \in \{0,1\}, \quad \forall j \end{aligned} \tag{4}$$

where Γ controls the strength of the robustness penalty.

We tested $\Gamma \in \{0.0, 0.01, 0.1, 1.0\}$ across sparsity levels $k \in \{5, 6, 7, 8, 9, 10\}$ to understand the interplay between sparsity and regularization strength. The optimal model used $k = 9$ features with $\Gamma = 1.0$. Interestingly, the features selected by the robust MIO model were identical to those from pure sparse MIO, suggesting that the optimal feature subset is stable across different regularization schemes. The ℓ_1 penalty provided marginal improvements in coefficient magnitudes but did not change the feature selection outcome.

3.1.4 Holistic Regression

Finally, we implemented **holistic regression**, a comprehensive optimization framework that simultaneously enforces multiple domain-informed constraints while exploring nonlinear feature transformations. This approach extends MIO sparse regression with additional constraint types:

1. Nonlinear Transformations: For each continuous feature x_j, we generated three additional candidate features: Squared: x_j^2, Square root: $\sqrt{x_j - \min(x_j) + \epsilon}$, and Logarithmic: $\log(x_j - \min(x_j) + 1)$ (captures exponential decay). This expanded the feature space from 19 to approximately 76 candidate features, allowing the model to capture nonlinear relationships observed in the residual plots (Figure 2).

2. Transformation Exclusivity: At most one transformation of each base feature can be selected:

$$z_j + z_{j,\text{sq}} + z_{j,\text{sqrt}} + z_{j,\log} \leq 1, \quad \forall j \tag{5}$$

This prevents redundancy by forcing the model to choose the single most informative representation of each base feature.

3. Group Sparsity (Clinical Domain Knowledge): We enforce that at most one feature from each clinically-related family can be selected:

$$\sum_{j \in G_{\text{jitter}}} z_j \leq 1, \qquad \sum_{j \in G_{\text{shimmer}}} z_j \leq 1, \qquad \sum_{j \in G_{\text{dynamics}}} z_j \leq 1 \tag{6}$$

where $G_{\text{jitter}} = \{\text{Jitter(\%)}, \text{Jitter(Abs)}, \text{Jitter:RAP}, ...\}$, $G_{\text{shimmer}} = \{\text{Shimmer}, \text{Shimmer(dB)}, ...\}$, and $G_{\text{dynamics}} = \{\text{RPDE}, \text{DFA}, \text{PPE}\}$. This constraint reflects clinical knowledge that features within each family measure similar underlying vocal properties and selecting multiple would be redundant.

4. Multicollinearity Control: Features with pairwise correlation $|\rho| \geq 0.8$ cannot both be selected:

$$z_i + z_j \leq 1, \quad \forall (i,j) : |\text{cor}(x_i, x_j)| \geq 0.8 \tag{7}$$

This ensures the final model avoids highly correlated features, improving coefficient stability and interpretability.

5. Statistical Significance: Selected features must have coefficients exceeding a minimum threshold $\beta_{\min} = 0.01$:

$$|\beta_j| \geq \beta_{\min} \cdot z_j, \quad \forall j \tag{8}$$

This is implemented using auxiliary binary variables and Big-M constraints to ensure that any selected feature has a meaningful contribution to the prediction.

The complete holistic regression formulation combines all constraints discussed above and sparsity and Big-M constraints:

$$\min_{\beta_0,\beta,z} \quad \sum_{i=1}^{n}(y_i - \beta_0 - \mathbf{x}_i^\top \boldsymbol{\beta})^2 + \Delta \sum_{j=1}^{p} |\beta_j| \tag{9}$$

We evaluated $k \in \{5, 6, 7, 8, 9, 10\}$ and $\Delta \in \{0.0, 0.1, 1.0\}$ with a time limit of 60 seconds per model due to the increased problem complexity. The optimal holistic model selected $k = 7$ features with $\Delta = 1.0$.

Table 1 summarizes the predictive performance and sparsity trade-offs across baseline linear models and optimization-based MIO sparse, robust, and holistic regressions. The features selected from these models are summarized in Table 2 below.

Model	**k**	**Γ**	**Test MAE**	**Test RMSE**
OLS (All features)	19	0.0	10.191	13.555
Lasso	16	varied	**9.967**	**13.415**
OLS (Lasso-selected features)	16	0.0	10.222	13.576
MIO Sparse	10	0.0	**9.907**	13.479
MIO Sparse + Robust	9	1.0	9.930	**13.419**
Holistic	7	1.0	12.075	15.023

Table 1: Performance Summary: Baselines vs. MIO Sparse/Robust/Holistic Models

3.2 Tree-Based Models

We next study a family of tree-based models implemented using the Interpretable AI (IAI) framework. All models are evaluated on the same four feature sets in order to isolate the effect of the learning algorithm from the choice of predictors. The first set reproduces the feature subset used in the original telemonitoring paper by Tsanas et al. (2010), and the remaining three are derived from our MIO–based sparse and robust feature selection, and from the holistic regression analysis in the previous section:

Set	**Description**	**Features**
Set 1	Paper Baseline (Tsanas et al.)	`["Jitter(Abs)", "Shimmer", "NHR", "HNR", "DFA", "PPE"]`
Set 2	MIO Sparse/Robust, $k = 10$	`["age", "test_time", "Jitter:DDP", "Shimmer:APQ5", "Shimmer:APQ11", "NHR", "HNR", "RPDE", "DFA", "PPE"]`
Set 3	MIO Sparse/Robust, $k = 9$	`["age", "test_time", "Shimmer:APQ5", "Shimmer:APQ11", "NHR", "HNR", "RPDE", "DFA", "PPE"]`
Set 4	Holistic Regression Features	`["HNR", "PPE", "age", "Jitter:DDP_log", "Shimmer:DDA_log"]`

Table 2: Feature Sets for Tree-Based Models

For each feature set, we train four tree-based methods: CART, Random Forests, Optimal Regression Trees (ORT), and XGBoost. Hyperparameters are selected by cross-validated mean squared error on the combined train+validation data, and final models are evaluated on the held-out test subjects. To preserve interpretability and avoid overly complex partitioning of the feature space, we restrict all tree-based models to relatively shallow trees. Concretely, we use:

- `max_depth` $\in \{2, 3, 4\}$: controls how many sequential splits a tree can make; we cap the depth so that the resulting decision paths remain short enough for clinicians to read and reason about.

- `minbucket` $\in \{30, 35, 40, 45, 50, 55, 60\}$: sets the minimum number of observations in each terminal node; this ensures that each leaf corresponds to a reasonably large group of patients rather than a handful of idiosyncratic cases.
- `num_trees` $\in \{20, 25, 50, 100\}$: determines the size of the Random Forest ensemble; we allow enough trees to stabilize predictions, while accepting that the resulting model is mainly used as a higher-accuracy benchmark rather than a fully interpretable tool.
- `num_estimators` $\in \{20, 25, 50, 100\}$: specifies the number of boosting iterations in XGBoost; this range lets the boosted model reach good predictive performance without pushing to extremely large ensembles, and we again view it primarily as a performance benchmark alongside the more interpretable single-tree models.

3.2.1 CART

As a baseline interpretable tree model, we fit CART using IAI's heuristic tree builder. CART greedily chooses splits by optimizing a local impurity reduction criterion (here, mean squared error) without attempting global optimality. In our clinical telemonitoring setting, this leads to a small tree that can be read as a set of simple "if–then" rules on voice features and demographics, which a clinician could plausibly inspect and use to understand how different patient profiles map to predicted UPDRS levels.

3.2.2 Random Forest

To improve predictive performance while retaining robustness to noise, we next consider Random Forests. In IAI, we use `RandomForestRegressor`, which aggregates an ensemble of decorrelated CART-style trees trained on bootstrap samples and random feature subsets at each split. Although this ensemble no longer yields a single global decision rule, the model can provide more accurate risk estimates for individual patients while still allowing partial interpretation through variable importance measures and representative trees. However, the overall forest remains substantially less interpretable than a single tree: a clinician cannot easily trace a patient-level prediction through hundreds of trees, so we view Random Forests primarily as a higher-accuracy but less transparent benchmark in our clinical setting.

3.2.3 Optimal Regression Trees

Optimal Regression Trees (ORT) aim to preserve the interpretability of a single decision tree while using mixed-integer optimization to search for a globally (or near-globally) optimal tree structure. Rather than committing to greedy split choices, ORT solves an optimization problem that balances fit to the data with a penalty on tree complexity, subject to depth and minimum leaf-size constraints. ORT demonstrates how discrete optimization can be used to systematically explore tree structures and potentially improve upon heuristic CART while remaining fully interpretable. In the clinical context, ORT helps obtain a compact set of decision rules that both perform competitively and can be read as transparent "if–then" pathways linking specific patterns of acoustic features and demographics to predicted UPDRS levels, which is crucial for interpreting model predictions and communicating the results to clinicians and patients.

3.2.4 XGBoost

Finally, we include a gradient-boosted tree model using IAI's `XGBoostRegressor`, which builds an additive ensemble of shallow regression trees via gradient boosting. This typically delivers the strongest predictive performance on tabular data, but at the cost of reduced interpretability compared to a single tree. In the clinical context, we therefore view XGBoost primarily as a performance benchmark: it shows how accurately UPDRS might be predicted from voice data if we are willing to sacrifice transparency, against which we can compare simpler models that provide clearer decision rules for patient care.

4 Results

4.1 Summary of Model Performace

Table 3 summarizes the test performance of the four tree-based methods across the four feature sets. Across all configurations, test MAE ranges from roughly 9 to 11 and RMSE from 12 to 15. Notably, the single-tree models (CART and especially ORT) often achieve better accuracy than the ensemble methods. Random

	CART		ORT		Random Forest		XGBoost	
Feature set	**MAE**	**RMSE**	**MAE**	**RMSE**	**MAE**	**RMSE**	**MAE**	**RMSE**
Set 1: paper baseline	10.176	14.345	10.008	14.418	10.146	14.141	10.222	13.888
Set 2: MIO Sparse/Robust	10.579	13.198	**9.111**	12.191	10.353	13.044	11.054	14.201
Set 3: MIO Sparse/Robust	10.579	13.198	**9.111**	12.191	10.353	13.044	10.991	14.491
Set 4: Holistic Regression	10.626	12.472	9.735	**12.177**	10.227	12.232	12.577	14.619

Table 3: Test MAE and RMSE for Tree-Based Models

Forest and XGBoost do not dominate despite their greater flexibility. The best linear/MIO model (MIO sparse + robust) attains MAE = 9.930 and RMSE = 13.419, which we treat as a strong regression baseline. Against this benchmark, the tree-based models, particularly ORT, reduce RMSE by about one UPDRS point and slightly lower the MAE, showing that they can capture useful nonlinear patterns in the features and improve accuracy without noticeably overfitting.

Within each feature set, ORT consistently dominates or matches the other tree methods. For example, on the MIO sparse/robust sets (Sets 2 and 3), ORT reaches MAE = 9.111 and RMSE = 12.191, which improves on both CART and the ensembles in terms of RMSE. This pattern shows the role of MIO in the linear setting: just as mixed-integer optimization finds better sparse linear predictors than simple Lasso or OLS subset selection, ORT uses optimization to find higher-quality trees than the greedy CART heuristic, while still yielding a single, interpretable model.

Comparing feature sets, we observe that the original paper subset (Set 1) is consistently weaker than the MIO-derived and holistic sets, echoing the improvement seen when moving from OLS with all features to MIO-selected subsets. Interestingly, the holistic feature set (Set 4) behaves very differently in the tree and linear worlds. As a linear model, the holistic specification performs poorly (MAE = 12.075, RMSE = 15.023), but when the same transformed predictors are fed into ORT, performance improves substantially to MAE = 9.735 and RMSE = 12.177, the best RMSE among all tree configurations. This indicates that, even after applying nonlinear transformations (logs) to stabilize scales and reduce curvature, a single global linear effect for each feature is still too restrictive. In practice, the impact of age, HNR, and PPE on UPDRS appears to change across different ranges of these variables, and some effects only show up for specific subgroups of patients. ORT can handle this behaviour by learning age- and feature-specific thresholds and interactions, effectively creating separate rules for different patient groups and better matching the observed patterns in the data.

4.2 Discussion

Taken together, these results motivate our choice of the **Set 4 ORT model as the main interpretable, well-performed model**: it leverages the holistic features in a piecewise nonlinear way to match or surpass the accuracy of both linear MIO models and more complex tree ensembles, while remaining sufficiently simple to inspect and explain in a clinical setting.

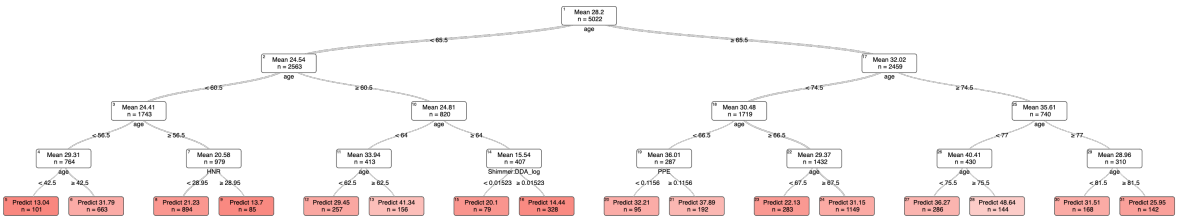

Figure 3: ORT Tree for Set 4 Holistic Features

Figure 3 shows the resulting optimal regression tree for Set 4. The tree remains shallow and interpretable.

All internal splits involve four features: age, HNR, PPE, and log-transformed shimmer DDA. The root node splits on age, effectively dividing patients into broad age bands, which then branch into subgroups based on voice regularity and noise (HNR and PPE) or amplitude perturbations (shimmer DDA). For example, on the left side of the tree, patients around 56–60 years old are further separated by HNR (harmonics-to-noise ratio, a measure of how periodic and "clean" the voice signal is). On the right side, older patients between roughly 66 and 74 years are split on PPE (pitch period entropy, which quantifies irregularity in pitch). This pattern suggests that, for younger patients, overall voice quality (HNR) is more informative for UPDRS, whereas for older patients, fine-grained pitch instability (PPE) becomes a stronger marker of disease severity, reflecting different speech characteristics in later-stage disease. Each leaf corresponds to a clinically interpretable subgroup (e.g., older patients with high PPE tend to have higher predicted UPDRS), and the associated mean prediction can be read directly from the node.

From a clinical perspective, this ORT model strikes a favorable balance between accuracy and transparency. It achieves error rates close to or better than more complex ensembles, while representing its decision logic as a small set of "if–then" rules linking age and a handful of acoustic markers to UPDRS severity. Such a structure can be inspected by clinicians, used to explain predictions to patients, and potentially inform hypotheses about how specific vocal changes relate to disease progression in Parkinson's disease.

5 Conclusion and Future Direction

Ultimately, to balance performance and interpretability, we recommend the ORT model with holistic regression features for clinical deployment. It achieved the best RMSE and competitive MAE using only 5 features selected through multicollinearity, sparsity, and robustness constraints. ORT outperformed complex ensembles (Random Forests, XGBoost), showing that with limited data, simpler models with stronger optimization maintain accuracy while preserving interpretability. The tree reveals younger patients' scores are driven by voice quality (HNR), while older patients show greater sensitivity to pitch irregularity (PPE).

Future work will aim to extend and refine the present analysis along several dimensions. First, the proposed models can be applied to larger and more heterogeneous cohorts, moving beyond the current sample of 42 early-stage subjects, in order to more rigorously assess external validity and robustness across patient populations. Second, a more detailed investigation of feature interactions is warranted, with a particular focus on identifying which acoustic patterns change most rapidly with disease progression and how these effects depend on age and other demographic covariates. Finally, it would be valuable to explore alternative optimal tree formulations and ensemble strategies within the IAI framework, with the goal of further improving predictive performance while retaining a high degree of interpretability for clinical use.

6 Contributions

This project represents an equal collaboration between Rebecca Chang and Ivy Xu, with regards to the code, report, and presentation. Rebecca implemented all optimization-based linear models, including OLS, Lasso, MIO sparse and robust regression, and the holistic regression framework. She conducted data preprocessing, feature engineering, and hyperparameter optimization for these approaches, and authored the corresponding Methods sections. Ivy implemented all tree-based models (CART, Optimal Regression Trees, Random Forests, and XGBoost) using the IAI framework. She performed hyperparameter tuning and cross-validation across all feature sets, generated the tree visualizations, and authored the tree-based Methods section and the Results section.

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